

1 Instantiate tasks



template

Buy a train ticket
from {from} to {to}
for {passenger}

from: Beijing

to: Shanghai

passenger: Mom



state patch

```
{"contact": "Mom",  
"messages": ["Please  
buy a ticket for  
me."]}
```

Patch into runtime overlay state

screenshots



Agent

Actions



initial
state

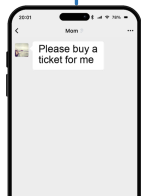
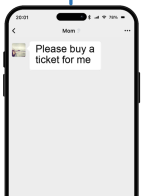
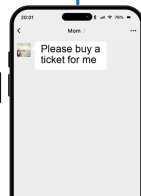
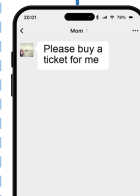
Structured State

```
{"user": {...}, "contacts": [...], "orders": [...], "settings": {...}}
```

clone
state

state diff

2 Fork parallel rollouts



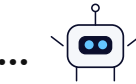
Rollout 1



Rollout 2



Rollout 3



Rollout N

3 Verify outcomes

State diff

```
+ tickets[0].status: "created"  
+ tickets[0].train_no: "G101"  
+ tickets[0].route: ["Beijing",  
"Shanghai"]  
+ tickets[0].passenger: "Mom"  
-----  
unexpected side effects:  
- order[0]; old order deleted
```



Judge

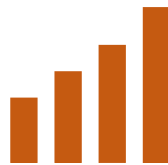


- ✓ ticket created
- ✓ route matches
- ✓ passenger matches
- ✗ no side effects

judge

4 Produce signals

Benchmark metrics



- ✓ SR: Task Success
- ✓ PR: Task Progress
- ✗ FC: False Completion
- ✗ USE: Side Effects
- ✗ OT: Overdue

RL reward

- ★ Success Reward (+1.0)
- ★ Progress Reward (dense)
- ✗ Side Effect Penalty
- ✗ False Completion Penalty